

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI

Semester: 5

Academic Year: 2026-27

Division: E

Batch: 3

Practical No: 2

Roll no: 05

PRN No: 12411040

Name of Student: **Prathmesh Patil**

Subject Name : Deep Learning

Problem Statement : Implement a Single-Layer Perceptron for binary classification and analyze the effect of weights, bias, and activation functions.

Algorithm :

START

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Import Required Libraries

(TensorFlow, NumPy, Matplotlib)

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Create Binary Classification

Dataset (AND Gate)

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Build Single-Layer Perceptron

(1 Dense Layer, 1 Neuron)

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Select Sigmoid Activation

Function

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Compile the Model

(Adam Optimizer + Binary Crossentropy)

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Train the Model

(Epochs = 500)

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Obtain Weights and Bias

|



Test Model on Dataset

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Display Predictions

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Analyze Weights, Bias &

Activation Function Effect

|



END

Code :

```
import tensorflow as tf

import numpy as np

import matplotlib.pyplot as plt

X = np.array([[0,0],
               [0,1],
               [1,0],
               [1,1]], dtype=float)

y = np.array([[0],
               [0],
               [0],
               [1]], dtype=float)

model = tf.keras.Sequential([
    tf.keras.layers.Dense(
        1,
        input_shape=(2,),
        activation='sigmoid' )
```

```
)  
  
model.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)  
  
history = model.fit(  
    X,  
    y,  
    epochs=500,  
    verbose=0  
)  
  
weights, bias = model.layers[0].get_weights()  
  
print("Weights:\n", weights)  
print("\nBias:\n", bias)  
predictions = model.predict(X)  
  
print("Predictions:")  
print(predictions)  
plt.plot(history.history['loss'])  
plt.title("Training Loss")  
plt.xlabel("Epoch")
```

```
plt.ylabel("Loss")
```

```
plt.show()
```

Output :

